

CS23532-COMPUTER NETWORKS-LAB MANUAL

Practical -1

AIM: - Study of various Network commands used in Linux and Windows:

Windows commands

```
C:\Users\Seeralan M>hostname
```

```
Seeralan M
```

```
C:\Users\Seeralan M>ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : Seeralan M
```

```
Primary Dns Suffix . . . . . :
```

```
Node Type . . . . . : Mixed
```

```
IP Routing Enabled. . . . . : No
```

```
WINS Proxy Enabled. . . . . : No
```

Ethernet adapter Ethernet 2:

```
Connection-specific DNS Suffix . :
```

```
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
```

```
Physical Address. . . . . : 0A-00-27-00-00-0E
```

```
DHCP Enabled. . . . . : No
```

```
Autoconfiguration Enabled . . . . : Yes
```

```
Link-local IPv6 Address . . . . . : fe80::aa4a:775:1de1:1757%14(Preferred)
```

```
IPv4 Address. . . . . : 192.168.56.1(Preferred)
```

```
Subnet Mask . . . . . : 255.255.255.0
```

Default Gateway :
DHCPv6 IAID : 118095911
DHCPv6 Client DUID. : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F
NetBIOS over Tcpip. : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. : F8-FE-5E-3A-70-56
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. : FA-FE-5E-3A-70-55
DHCP Enabled. : No
Autoconfiguration Enabled : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description : Intel(R) Wi-Fi 6E AX211 160MHz
Physical Address. : F8-FE-5E-3A-70-55

DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes
Link-local IPv6 Address : fe80::b810:29c9:55e9:3a65%17(Preferred)
IPv4 Address. : 10.83.96.83(Preferred)
Subnet Mask : 255.255.255.0
Lease Obtained. : 26 August 2026 14:38:14
Lease Expires : 26 August 2026 15:38:11
Default Gateway : 10.83.96.20
DHCP Server : 10.83.96.20
DHCPv6 IAID : 167312990
DHCPv6 Client DUID. : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F
DNS Servers : 10.83.96.20
NetBIOS over Tcpi. : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Bluetooth Device (Personal Area Network)
Physical Address. : F8-FE-5E-3A-70-59
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Realtek Gaming GbE Family Controller

Physical Address. : 2C-58-B9-DE-5F-5F
DHCP Enabled. : Yes
Autoconfiguration Enabled . . . : Yes

C:\Users\Seeralan M>arp -a

Interface: 192.168.56.1 --- 0xe

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 10.83.96.83 --- 0x11

Internet Address	Physical Address	Type
10.83.96.20	4e-ec-85-ec-ff-15	dynamic
10.83.96.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

C:\Users\Seeralan M>netstat -r

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Interface List

14...0a 00 27 00 00 0eVirtualBox Host-Only Ethernet Adapter
11...f8 fe 5e 3a 70 56Microsoft Wi-Fi Direct Virtual Adapter
4...fa fe 5e 3a 70 55Microsoft Wi-Fi Direct Virtual Adapter #2
17...f8 fe 5e 3a 70 55Intel(R) Wi-Fi 6E AX211 160MHz
19...f8 fe 5e 3a 70 59Bluetooth Device (Personal Area Network)
2...2c 58 b9 de 5f 5fRealtek Gaming GbE Family Controller
1.....Software Loopback Interface 1

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IPv4 Route Table

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Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	10.83.96.20	10.83.96.83	35	
10.83.96.0	255.255.255.0	On-link	10.83.96.83	291	
10.83.96.83	255.255.255.255	On-link	10.83.96.83	291	
10.83.96.255	255.255.255.255	On-link	10.83.96.83	291	
127.0.0.0	255.0.0.0	On-link	127.0.0.1	331	
127.0.0.1	255.255.255.255	On-link	127.0.0.1	331	
127.255.255.255	255.255.255.255	On-link	127.0.0.1	331	
192.168.56.0	255.255.255.0	On-link	192.168.56.1	281	
192.168.56.1	255.255.255.255	On-link	192.168.56.1	281	
192.168.56.255	255.255.255.255	On-link	192.168.56.1	281	
224.0.0.0	240.0.0.0	On-link	127.0.0.1	331	
224.0.0.0	240.0.0.0	On-link	192.168.56.1	281	
224.0.0.0	240.0.0.0	On-link	10.83.96.83	291	
255.255.255.255	255.255.255.255	On-link	127.0.0.1	331	
255.255.255.255	255.255.255.255	On-link	192.168.56.1	281	

255.255.255.255 255.255.255.255 On-link 10.83.96.83 291

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Persistent Routes:

None

IPv6 Route Table

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Active Routes:

If	Metric	Network	Destination	Gateway
1	331	::1/128	On-link	
14	281	fe80::/64	On-link	
17	291	fe80::/64	On-link	
14	281	fe80::aa4a:775:1de1:1757/128		
			On-link	
17	291	fe80::b810:29c9:55e9:3a65/128		
			On-link	
1	331	ff00::/8	On-link	
14	281	ff00::/8	On-link	
17	291	ff00::/8	On-link	

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Persistent Routes:

None

C:\Users\Seeralan m>nslookup www.google.com

Server: UnKnown

Address: 10.83.96.20

Non-authoritative answer:

Name: www.google.com

Addresses: 2001:4860:482d:7700::

2001:4860:4826:7700::

2001:4860:4828:7700::

2001:4860:482b:7700::

2001:4860:4827:7700::

2001:4860:4829:7700::

2001:4860:482a:7700::

2001:4860:482c:7700::

142.251.152.119

142.251.156.119

142.251.154.119

142.251.151.119

142.251.153.119

142.251.157.119

142.251.150.119

142.251.155.119

C:\Users\Seeralan M>ping google.com

Pinging google.com [142.251.43.46] with 32 bytes of data: Reply

from 142.251.43.46: bytes=32 time=14ms TTL=114

Reply from 142.251.43.46: bytes=32 time=233ms TTL=114

Reply from 142.251.43.46: bytes=32 time=59ms TTL=114

Reply from 142.251.43.46: bytes=32 time=328ms TTL=114

Ping statistics for 142.251.43.46:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 14ms, Maximum = 328ms, Average = 158ms

C:\Users\Seeralan M>route print

Interface List

14...0a 00 27 00 00 0eVirtualBox Host-Only Ethernet Adapter
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4...fa fe 5e 3a 70 55Microsoft Wi-Fi Direct Virtual Adapter #2
17...f8 fe 5e 3a 70 55Intel(R) Wi-Fi 6E AX211 160MHz
19...f8 fe 5e 3a 70 59Bluetooth Device (Personal Area Network)
2...2c 58 b9 de 5f 5fRealtek Gaming GbE Family Controller
1.....Software Loopback Interface 1

IPv4 Route Table

Active Routes:

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	10.83.96.20	10.83.96.83	35
10.83.96.0	255.255.255.0	On-link	10.83.96.83	291
10.83.96.83	255.255.255.255	On-link	10.83.96.83	291
10.83.96.255	255.255.255.255	On-link	10.83.96.83	291
127.0.0.0	255.0.0.0	On-link	127.0.0.1	331
127.0.0.1	255.255.255.255	On-link	127.0.0.1	331
127.255.255.255	255.255.255.255	On-link	127.0.0.1	331
192.168.56.0	255.255.255.0	On-link	192.168.56.1	281

192.168.56.1	255.255.255.255	On-link	192.168.56.1	281
192.168.56.255	255.255.255.255	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	127.0.0.1	331
224.0.0.0	240.0.0.0	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	10.83.96.83	291
255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
255.255.255.255	255.255.255.255	On-link	192.168.56.1	281
255.255.255.255	255.255.255.255	On-link	10.83.96.83	291

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Persistent Routes:

None

IPv6 Route Table

Active Routes:

If	Metric	Network	Destination	Gateway
1	331	::1/128		On-link
14	281	fe80::/64		On-link
17	291	fe80::/64		On-link
14	281	fe80::aa4a:775:1de1:1757/128		
				On-link
17	291	fe80::b810:29c9:55e9:3a65/128		
				On-link
1	331	ff00::/8		On-link
14	281	ff00::/8		On-link
17	291	ff00::/8		On-link

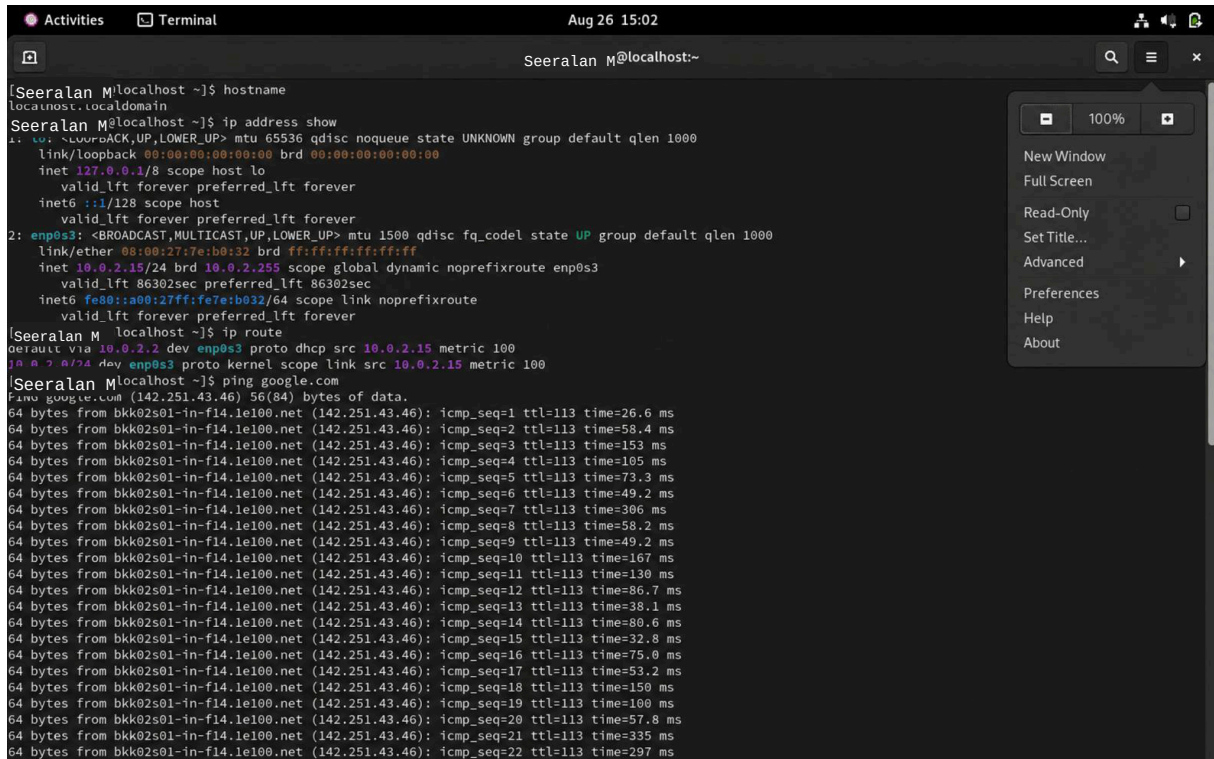
=====

Persistent Routes:

None

C:\Users\Seeralan M>tracert google.com

Linux commands



The screenshot shows a Linux terminal window titled "Seeralan M@localhost:~" with a date and time of "Aug 26 15:02". The terminal displays the output of several commands:
1. `hostname`: Returns `localhost.localdomain`.
2. `ip address show`: Shows details for the `lo` loopback interface (127.0.0.1) and the `enp0s3` ethernet interface (10.0.2.15).
3. `ip route`: Shows the default route via `10.0.2.2` on `enp0s3`.
4. `ping google.com`: Shows 22 successful ping attempts to `142.251.43.46` with varying response times between 26.6 ms and 335 ms.
On the right side of the terminal window, a context menu is open, showing options like "New Window", "Full Screen", "Read-Only", "Set Title...", "Advanced", "Preferences", "Help", and "About".

Student observation answers

1. **Command used to check reachability:** ping
2. **Command used to find hops:** tracert in Windows / traceroute in Linux
3. **Command to display IP configuration:** ipconfig in Windows / ip address in Linux
4. **Command to display TCP port status:** netstat
5. **Command to modify IP configuration in Linux:** ip address / ip route